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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

CS3222-1 – CRYPTOGRAPHY AND NETWORK SECURITY

(For CS)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following ciphers are created by shuffling the letters of a word?
A) substitution cipher B) transposition cipher
C) vigenere cipher D) hill cipher
2. Cryptanalysis is used _____
A) to make new ciphers B) to encrypt the data
C) to find some insecurity in a cryptographic scheme D) to increase the speed
3. In the RSA algorithm, we select 2 random large values 'p' and 'q'. Which of the following is the property of 'p' and 'q'?
A) p and q should be divisible by $\Phi(n)$ B) p and q should be co-prime
C) p and q should be prime D) p/q should give no remainder
4. Which of the following statements are true?
i) Stream Ciphers are faster than Block Ciphers
ii) Block Ciphers can reuse keys
iii) Block ciphers use lesser code than stream ciphers
A) 1st only B) 2nd and 3rd
C) 1st and 2nd D) All are true
5. For p = 11 and q = 19 and choose e=17. Apply RSA algorithm where message=5 and find the cipher text.
A) C=92 B) C=80
C) C=56 D) C=23
6. In the DES algorithm the round key is _____ bit and the Round Input is _____ bits.
A) 56, 24 B) 64, 32
C) 32, 32 D) 48, 32
7. Caesar Cipher is an example of
A) Poly-alphabetic Cipher B) Mono-alphabetic Cipher
C) Multi-alphabetic Cipher D) Bi-alphabetic Cipher
8. What is the primary goal of a Man-in-the-Middle (MITM) attack?
A) Unauthorized access to sensitive information B) Encryption of files for ransom
C) Intercepting and manipulating communication between two parties D) Disrupting network services

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Demonstrate the model of symmetric cryptosystem.	8	L*3	1	1
	b) Explain Playfair cipher with an example.	8	L2	1	1
2.	a) Construct the single round of DES algorithm.	8	L3	2	2
	b) Illustrate various cryptanalysis attacks based on information known to cryptanalyst.	8	L2	1	2
3.	a) Demonstrate RSA algorithm with example.	8	L3	2	2
	b) Explain Hill cipher with example.	8	L2	1	1
Unit – II					
4.	a) Demonstrate how diffie hellman key exchange protocol is vulnerable to man in middle attack.	8	L3	3	1
	b) Illustrate the key distribution scenario in Symmetric key distribution using Symmetric encryption.	8	L2	4	1
5.	a) Develop diffie hellman key exchange algorithm with example.	8	L3	3	2
	b) Explain the kerberose version 4 message exchange.	8	L2	4	2
6.	a) Construct general format of X509 certificate.	8	L3	4	3
	b) Illustrate the Elgamal Cryptographic systems with example.	8	L2	3	1
Unit – III					
7.	a) Interpret web security threats in network security.	8	L2	5	1
	b) Develop SSL architecture.	8	L3	5	3
8.	a) Explain in detail the basic combinations of security associations.	8	L2	5	1
	b) Demonstrate IP traffic processing model for Outbound packets.	8	L3	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
